

Helix OTA “ Issues (open workable items)

Revision: 8 **Last modified:** 2026-06-21T18:00:00Z

This is the canonical open-work tracker (Â§11.4.15 Status, Â§11.4.16 Type, Â§11.4.54 OTA-NNN). Closed items migrate to [Fixed.md](#). The short-form companion is [Issues_Summary.md](#).

Â§3. [OTA-003] Emulator Tier-2 “ real Android A/B (update_engine/AVB/dm-verity auto-rollback)

Status: In testing **Type:** Task

Description: Stand up the Tier-2 emulator “ a real Android A/B update_engine payload-apply with AVB/dm-verity verification and auto-rollback, driven end-to-end against the control plane (per docs/design/EMULATED_DEVICE_TESTING.md). The Android emulator (API 36, CZ_API36_Phone, Android 16) is now running on nezha.local (Linux x86_64 host, 62 GB RAM, KVM) and is reachable via ADB at emulator-5554. The HelixTrack API is accessible from the emulator via SSH tunnel. Cuttlefish (cvd) Tier-2 remains pending AOSP guest images.

Â§4. [OTA-004] Emulator Tier-3 “ real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions

Status: Operator-blocked **Type:** Task

Operator-Block-Details: - **WHAT:** Tier-3 validation on the physical board “ real vendor HAL, U-Boot A/B slot-switch, and dm-verity over real partitions, exactly as a fielded device runs (per docs/design/EMULATED_DEVICE_TESTING.md). - **WHY:** Requires the physical RK3588 / Orange Pi 5 Max hardware, which is not attached to this host. Self-resolution exhausted (Â§11.4.21): (a) no remote board over the available CLIs/ADB/SSH; (b) subagent delegation cannot substitute for absent silicon; (c) no repo tooling emulates real U-Boot slot-switch + on-silicon dm-verity; (d) no captured fallback reproduces real-partition AVB rollback; (e) research confirms the vendor HAL + bootloader path is board-specific. NOT structurally impossible (Â§11.4.112) “ hardware-gated only. - **UNBLOCK CONDITION:** A physical RK3588 / Orange Pi 5 Max board reachable over ADB/SSH (or physically attached) for flashing. - **WHO:** Operator “ attach / provide remote access to the board.

Â§5. [OTA-021] HelixTrack bidirectional sync verification

Status: In progress **Type:** Task

Description: Verify that changes to workable items docs/DB are immediately synced to HelixTrack tickets and vice versa. End-to-end sync test with recorded video evidence.
